

Luke Slater

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EDUCATION

Bachelor of Science in Mechanical Engineering

Minor in Computer Science

Western-CU Boulder Engineering Partnership Program

GPA: 3.75

Gunnison, CO

Graduation Date: May 2025

SOFTWARE SKILLS

- SolidWorks
- C++
- Excel
- Python “Snake Charmer”
- Kali Linux
- Microsoft Project
- GitHub

FABRICATION SKILLS

- MIG Welding
- Lincoln CNC Plasma Table
- HAAS Mini Mill
- Sharp Precision Lathe
- Sharp Vertical Manual Mill
- Epilog Fusion 48 Laser

ENGINEERING EXPERIENCE

Ultralight Adaptive Gravel Bike Project

University of Colorado Boulder / Adaptive Sports Center of Crested Butte

08/2023 – 05/2024

- Designed main steering assembly incorporating titanium lugs coupled with carbon tubing, enabling rear-wheel drive, and achieving a lightweight assembly.
- Implemented effective communication channels, enhancing the team's overall organization.
- Ensured the team's file-sharing software was functioning properly, avoiding conflicting file versions.
- Contributed to the development of formal write-up, emphasizing part drawing.

Mountain Bike Wheel Testing

Blister Review

Summer 2024

- Developed Python code to analyze mountain bike wheel deformations using motion capture data
- Extrapolated deflection values from captured data
- Employed quaternions and matrix transformations for precise analysis
- Collaborated with Blister to publish the complex findings in a clear and concise manner

Ski Flex Tester

Blister Review

Summer 2023

- Formulated mathematical equations to guarantee accurate readings, then integrated them into MATLAB code.
- Precision machined complex parts from expensive stock, crucial for critical components. This responsibility cultivated trust among my team and supervisors, establishing confidence in handling other complex machining tasks.
- Prototyped specific components on the fly, improving the machines' functionality and eliminating outlier data.

Human Movement Project

University of Colorado Boulder

01/2024 - 05/2024

- Conducted experiments to test the hypothesis that electromyography (EMG) data could accurately identify deliberate attempts to disrupt balance, even when such attempts were not visually apparent.
- Developed MATLAB code to process raw EMG data into power spectral density (PSD) data, incorporating ninety percent confidence intervals. This analysis revealed that intentional attempts to disrupt balance were associated with significantly increased muscle activation in the peroneus longus.

Ski Engineering and Design

University of Colorado Boulder

08/2024 – Current

- Designed and manufactured custom skis with tailored performance characteristics, utilizing materials such as epoxy, wood, and plastic.
- Programmed G-code for CNC router operations to precisely cut ski cores, bases, and sidewalls, enabling complex tooling and custom design execution.

LEADERSHIP EXPERIENCE

Senior Capstone Project Manager | Western-CU Boulder Engineering Partnership Program | Gunnison, CO | Sep. 2024 -Present

- Assigned project tasks, maintained and updated the team's Gantt chart.
- Facilitated team meetings and ensured clear communication with project sponsors and stakeholders.

High School Leader | YoungLife | Crested Butte, CO | Sep. 2022 -Present

- Demonstrated leadership in fostering a sense of community and shared values among high school students.

STUDENT AFFILIATIONS

American Society of Mechanical Engineers, Gunnison, CO

8/2023-Present